# Exercise 1: Map the Roman Empire

**Dataset:** Roman Empire Infrastructure

In this exercise you make two maps of the Roman Empire and a print layout with both maps. You practise loading raster and vector data, layer order, raster styling, categorized and graduated styles, labels, pie charts, map themes, coordinate systems and print layouts.

## A) Map the empire and its landscape

*You can do this after chapter 4.*

1. Load all layers of the dataset. Put them in a logical order.
2. Show `elevation`: low land in green, mountains in brown or red. Use *Identify* to find the elevation of a city in the Alps.
3. Show the provinces with an outline only, so that the elevation stays visible.
4. Show the roads with a categorized style: major roads thicker than minor roads.
5. Show the cities with a categorized style on `type`: cities as small dots, legionary fortresses as squares. Label the cities with their `name`.
6. Save the current styles as a map theme.

## B) Map the infrastructure per province

*You can do this after chapter 4.*

1. Restyle the provinces. Make sure you don't lose your previous style: first create a new style, then start configuring.
2. Colour the provinces with a graduated style on `area_km2`: the larger the province, the darker the colour. Label the provinces with their `name`.
3. Add a pie chart to every province with the number of monuments of each type (`n_theatres`, `n_amphitheatres`, `n_circuses`, `n_baths`, `n_temples`). The more monuments in total (`n_monuments`), the larger the pie chart.
4. Save the current styles as a map theme. Switch between the two map themes to check that both maps are kept.

## C) Prepare for finishing

*You can do this after chapter 5.*

1. Set the project to a projected CRS that suits the empire and compare the shapes of the provinces before and after.
2. Save the project in the folder of the dataset. Keep it: you need it in the next step.

## D) Make a print layout

*You can do this after chapter 6.*

Make a finished map, as a PDF, with:

1. An overview map of the whole empire, showing the map you made in part B (provinces and pie charts). This is the largest map on the page. It shows a frame with the extent of the detail map.
2. A detail map that zooms in on the province of Belgica, showing the map you made in part A (elevation, roads and cities).

3. A title, and a legend, a scale bar and a north arrow that are linked to the overview map.